

URONIC ACID PATHWAY

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Objectives:

- At the end of this lecture, it is expected to;
 - Describe the biomedical importance of uronic acid pathway
 - Describe the reactions of uronic acid pathway
 - Explain the clinical importance of uronic acid pathway
 - Essential pentosuria
 - Oxalosis

Biomedical Importance of Uronic Acid Pathway:

- Major function is to produce D-Glucuronic acid from glucose:
 - For excretion of metabolites and foreign chemicals (xenobiotics) as **glucuronides**.
 - For e.g., Bilirubin is **conjugated** with Glucouronic acid to produce bilirubin diglucuronides, which is water soluble and readily excreted through bile.
 - UDP-glucuronate is a precursor in biosynthesis of proteoglycans and glycoproteins.
- Energy is not produced in this pathway.
- A source of UDP-glucose, which is used for glycogen formation.
- Provides a mechanism by which dietary D-xylulose can enter the central metabolic pathway.
- A deficiency in this pathway (Xylitol dehydrogenase/ Dicarboxyl and L-xylulose reductase (DCXR) deficiency) □ **Essential pentosuria**.
- In lower animals (not in primates), this pathway leads to synthesis of Vit C.
 - Due to lack of one enzyme of the pathway (L-gulonolactone oxidase), humans (primates and some other animals) don't synthesis **ascorbic acid** and should take through diet.

Conjugation of bilirubin with glucuronic acid

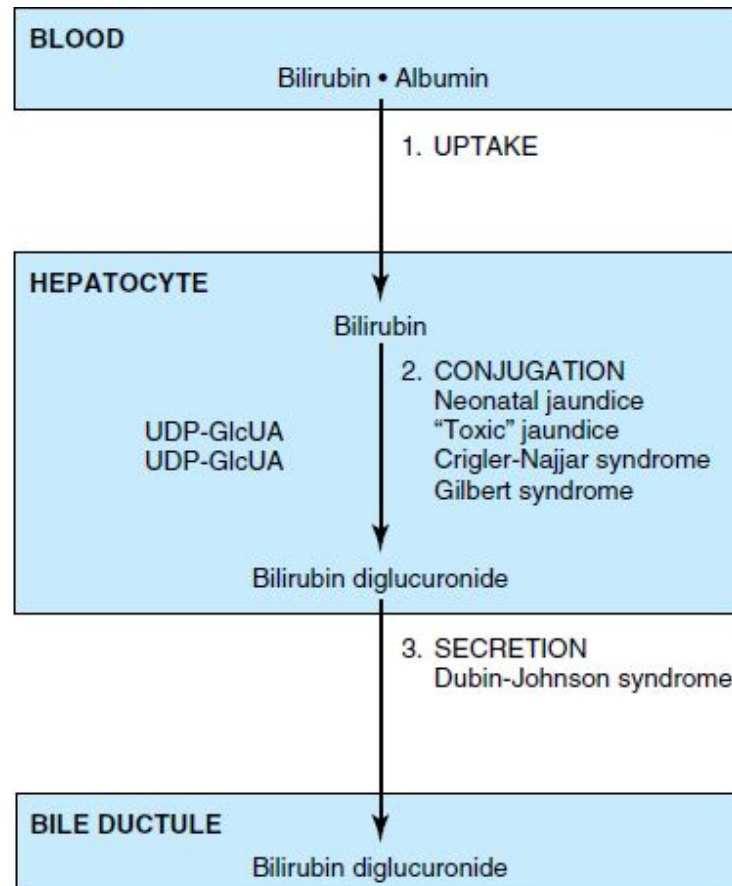
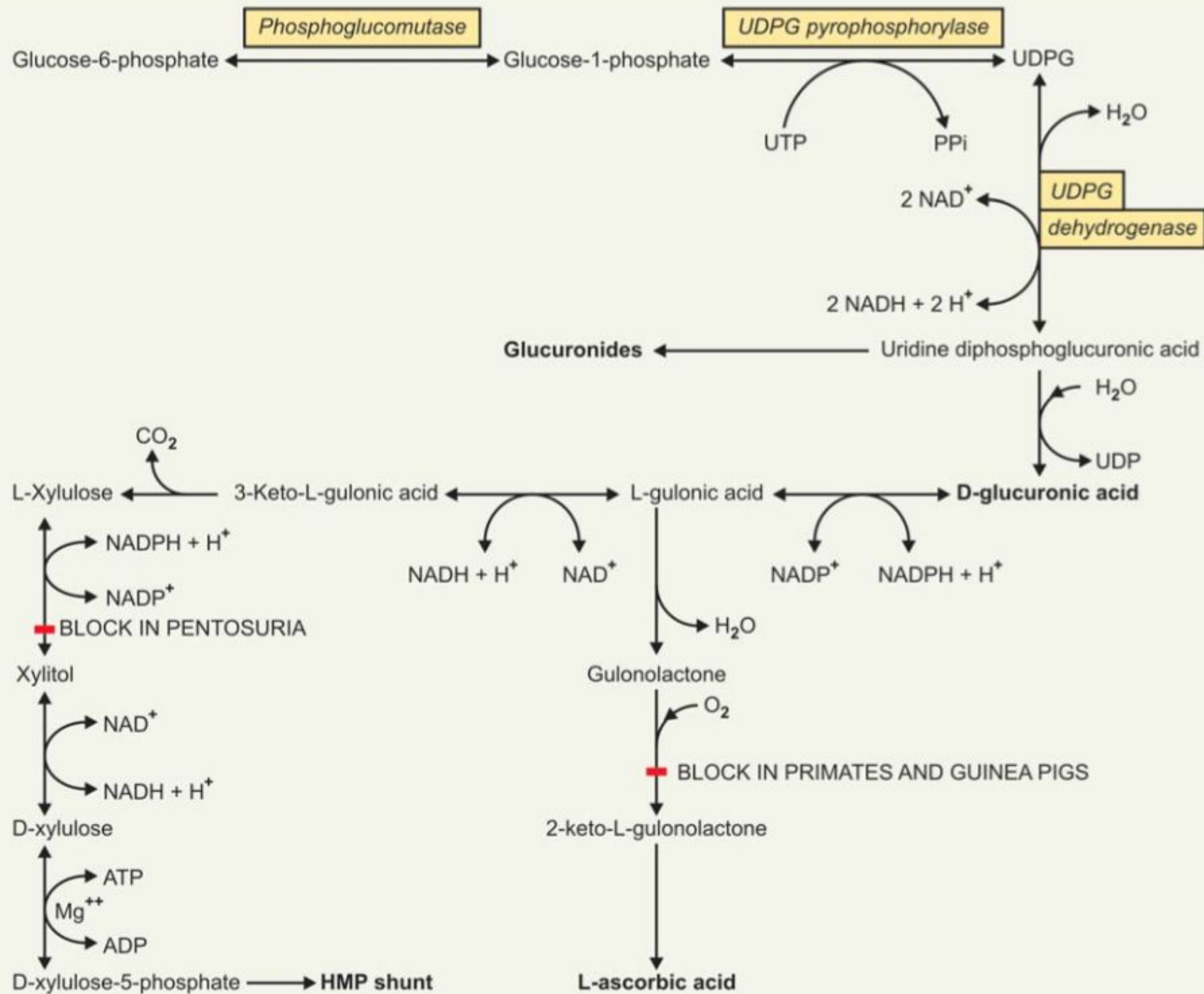


Figure 32–15. Diagrammatic representation of the three major processes (uptake, conjugation, and secretion) involved in the transfer of bilirubin from blood to bile. Certain proteins of hepatocytes, such as ligandin (a family of glutathione S-transferase) and Y protein, bind intracellular bilirubin and may prevent its efflux into the blood stream. The process affected in a number of conditions causing jaundice is also shown.

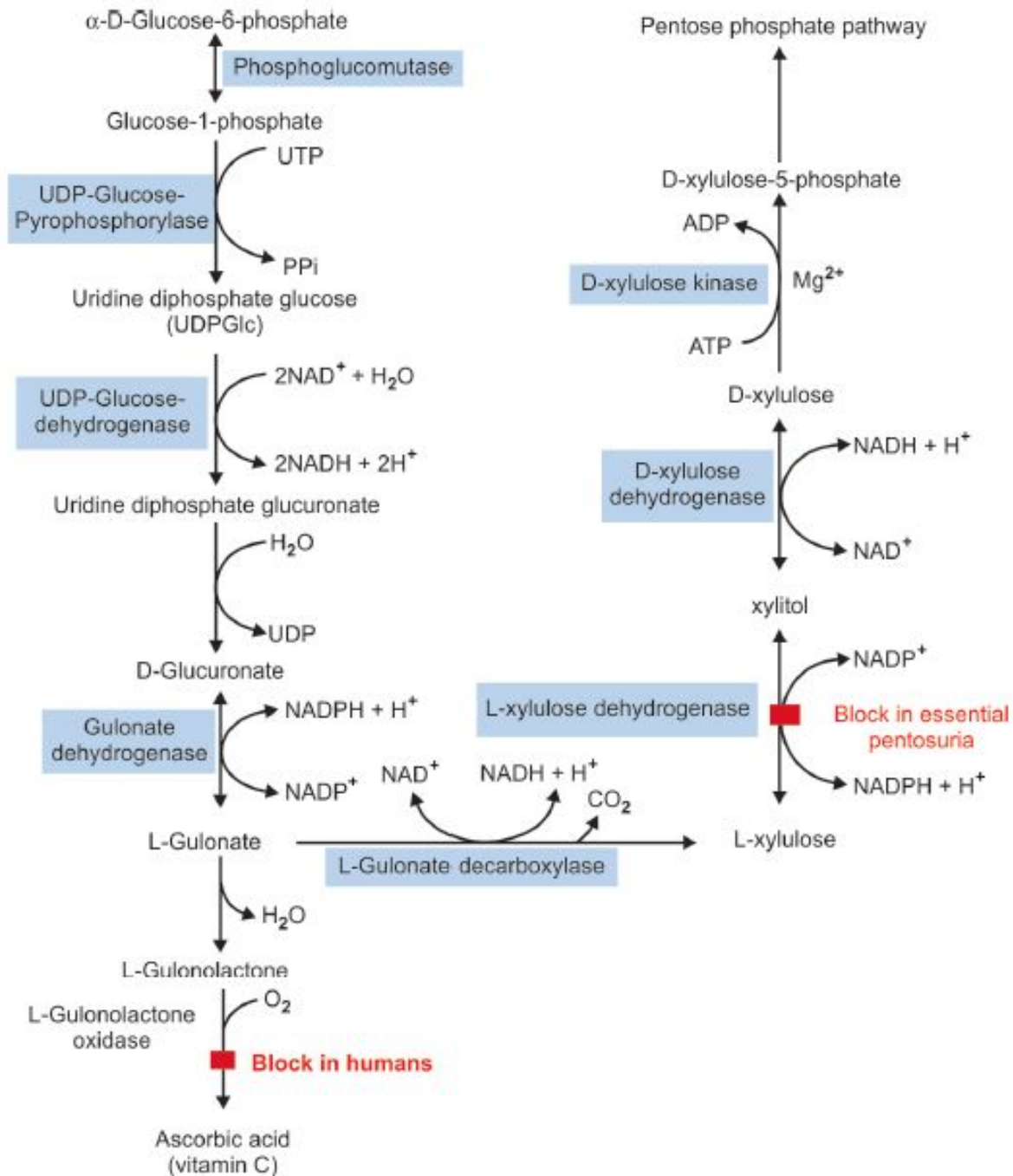
Uronic acid pathway:

- An alternative oxidative pathway for glucose.
- Takes place in Liver (cytoplasm)
- Convert glucose to glucuronic acid, ascorbic acid (not in human), and pentoses
- Like the pentose phosphate pathway—it does not lead to the generation of ATP.
- Source of glucuronic acid for conjugation of many endo- and exotoxins before excretion as glucuronides in urine and bile.

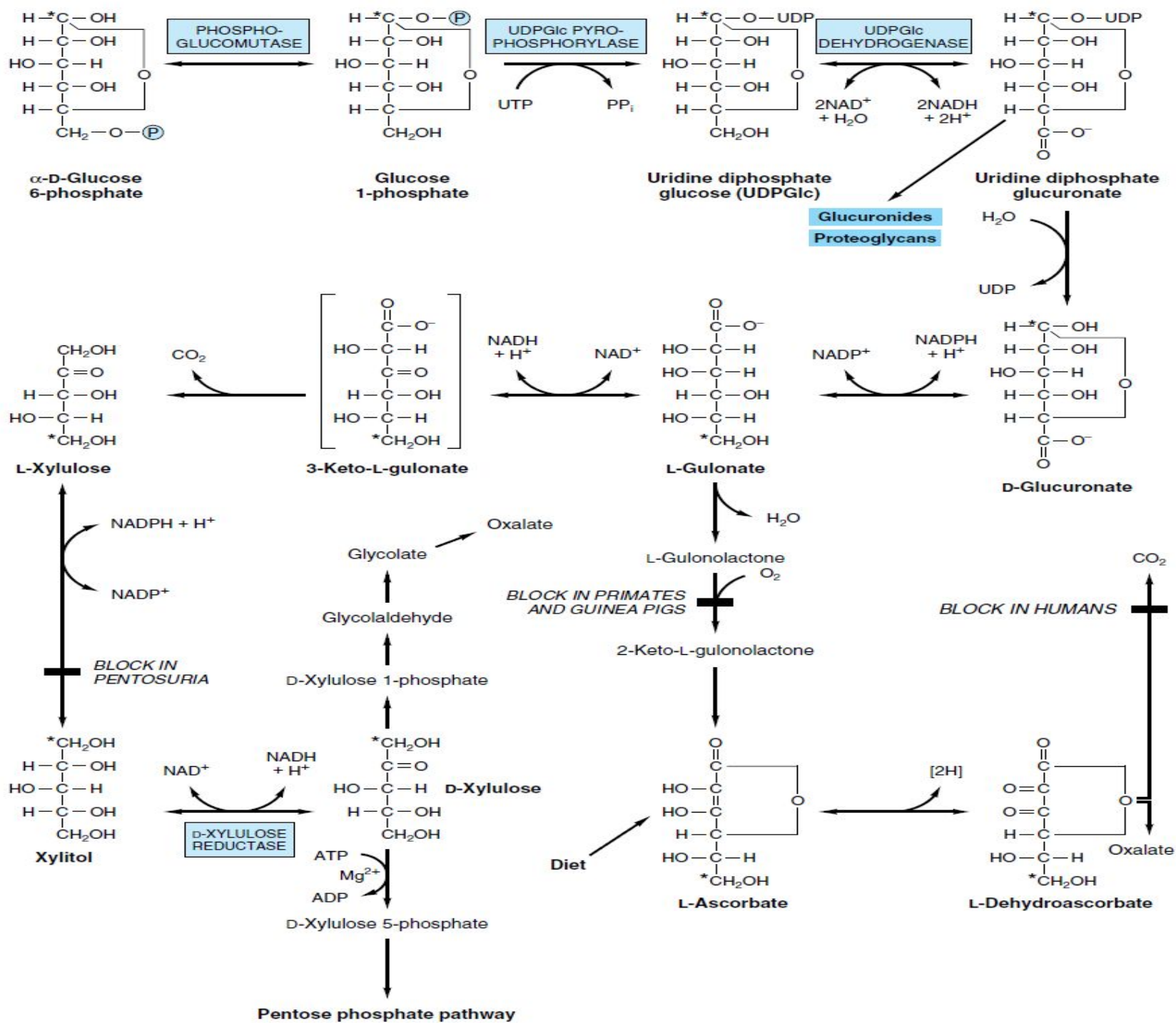
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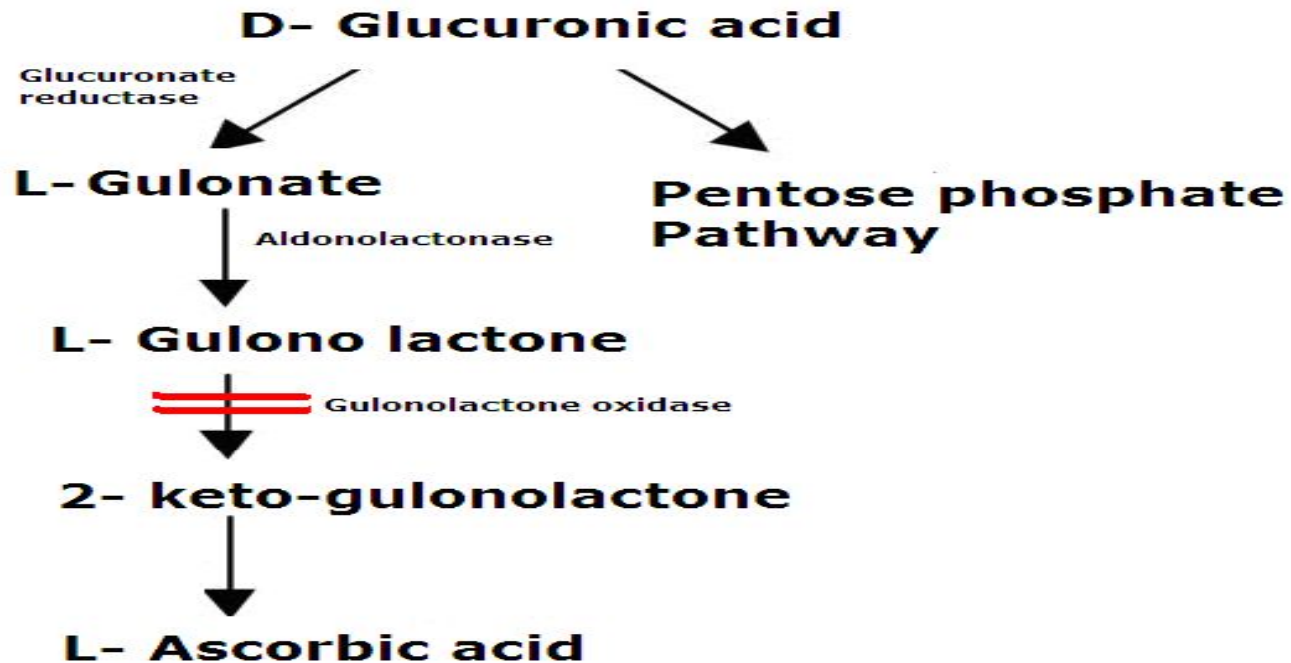


URONIC ACID PATHWAY



Relations with HMP SHUNT

- The unutilized Glucuronate produced in this pathway is converted to Xylulose-5 P which is further metabolized through HMP pathway



CLINICAL IMPORTANCE OF URONIC ACID PATHWAY

- Disruption of the Uronic Acid Pathway Caused by Enzyme Defects & Some Drugs - Results in defective xenobiotics metabolism and its excretion.
 - Essential pentosuria (enzyme deficiency, see in the next slide)
- Various drugs markedly increase the rate at which glucose enters the uronic acid pathway:
 - Administration of drugs i.e., Chlorobutanol & Barbitol significantly increases the uronic acid pathway.
- Certain drugs are also found to enhance the synthesis of Ascorbic acid.
 - Administration of aminopyrine and antipyrine were found to enhance the synthesis of ascorbic acid in rats

Essential pentosuria:

- Rare hereditary disease (Autosomal recessive disorder; DCXR gene; 17q25.3)
- Asymptomatic; No ill-effects
- One of Garrod's tetrad [alkaptonuria, albinism, pentosuria, cystinuria - inborn error of metabolism].
- 1 in 2500 births
- Metabolic defect:
 - Deficiency of dicarbonyl and L-xylulose reductase - necessary to reduce L-xylulose to xylitol.
- Considerable quantities of **L-xylulose** appear in the urine
- +ve benedict's test;
 - Differentiated from Diabetes Mellitus by +ve Bials test (for pentose sugars).

Oxalosis

- Primary hyperoxaluria Type I:
 - Rare metabolic disorder (An autosomal recessive disorder; AGXT gene; 2q37.3)
 - Due to deficiency of enzyme alanine-glyoxylate aminotransferase (Primary hyperoxaluria)
 - Found in peroxisomes in liver – converts glyoxylate to glycine.
 - Oxalosis occurs when kidney stop eliminating calcium oxalate from the body.
 - Oxalate crystals deposit elsewhere in the body.
 - Parenteral administration of xylitol may involving calcium oxalate deposition in brain and kidneys – lead to **oxalosis**.

Read more...



THANK YOU...